



**SAVEETHA
ENGINEERING COLLEGE**

Affiliated to Anna University | Approved by AICTE

AUTONOMOUS



Sl. No.: 19A1406

Marks	
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CIA-III Examination - April/May/Nov/Dec 2023

Register No.: 212222230134

Branch & Sem.: B-Tech AI-DS

Date: 21/11/23

Sign of the Invigilator:

Subject Code & Subject Name: 19A1406 ① Digital Image Processing

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PART-B

6A) $A = \{k_1, k_2, k_3, k_4, k_5\}$ with prob. probability $P(k_1) = P(k_3) = 0.2$
 $P(k_2) = 0.4, P(k_5) = 0.1$

- Find Huffman code
- average length and its redundancy?
- efficiency

7. $A = \{a_1, a_2, a_3, a_4, a_5\}$ with prob. probability $P(a_1) = 0.3$
 $P(a_2) = 0.4, P(a_3) = 0.15, P(a_4) = 0.05$ and $P(a_5) = 0.1$

- Huffman code
- avg length and redundancy
- Efficiency.

8. B.

PART-C

$r_0 = 1008, r_1 = 320, r_2 = 456, r_3 = 686, r_4 = 803, r_5 = 105$
 $r_6 = 417, r_7 = 301$

64×64 image.